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Artificial intelligence technologies in the management of environmental educational projects: A pedagogical experiment

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Abstract Despite growing empirical research on artificial intelligence (AI) in education, studies specifically addressing AI-driven management of environmental educational projects in higher education institutions (HEIs) remain scarce. This study addresses that gap by developing and validating the Integrated AI-driven Environmental Educational Project Management Framework (IAEPMF) – a four-level model combining adaptive machine learning (ML), generative AI, Retrieval-Augmented Generation (RAG), and cloud platforms to personalize and optimize environmental educational project workflows. A controlled pedagogical experiment was conducted over 18 weeks (2025) at three Ukrainian HEIs involving 120 students from environmental specialities (101 “Ecology”, 014.05 “Secondary Education: Biology and Human Health”, 017 “Physical Culture and Sports”). Participants were assigned to a control group (n=62, traditional instruction) and an experimental group (n=58, IAEPMF-based instruction). Data were collected through diagnostic tests, structured knowledge assessments, project evaluations, and student surveys covering nine performance indicators: academic performance, environmental competencies, task-completion speed, project work quality, learning self-reliance, motivation, digital literacy, critical thinking, and group work skills. Statistical analysis applied Student’s t-test and descriptive statistics using IBM SPSS Statistics 27. The experimental group demonstrated statistically significant improvements across all nine indicators ($p<0,01$). Academic performance increased by 25,0% (t=13,24), environmental competencies by 24,6% (t=10,78), digital literacy by 25,5% (t=6,45), group work skills by 24,7% (t=8,54), and critical thinking by 16,4% (t=6,71). The quality success rate rose from 34,5% to 60,3% in the experimental group, compared with 32,3% to 35,5% in the control group. The IAEPMF is confirmed as an effective, empirically validated instrument for transforming environmental higher education through AI-driven project management. The study identifies key systemic challenges and provides practical recommendations on phased integration, interdisciplinary team formation, external funding, faculty professional development, and cybersecurity compliance, contributing to the global evidence base for sustainable digital transformation in higher education.

Keywords: IAEPMF, machine learning, RAG, higher education institutions, pedagogical experiment

1. Introduction

Educational project management can be characterized as a pedagogical system of coordinated teaching and educational actions aimed at achieving educational, scientific, and formative goals within defined time and resource constraints using information and technical tools Oleksenko et al. (2025), Akour and Alenezi (2022). Within the framework of environmental education, such a system encompasses a series of organizational and managerial measures, such as: making IT-related decisions, planning the student learning processes, organizing scientific and pedagogical activities and environmental experiments, automating routine repetitive processes, motivating and providing feedback, regulating and coordinating the work of participants in the educational process, and finally, evaluating student learning outcomes Cherep et al. (2024), Goralski and Tan (2020).

Artificial intelligence in educational systems is implemented through machine learning algorithms — based on studying similarity, simulation, modeling, and identifying cause-and-effect relationships in educational processes and phenomena — natural language processing systems (text, audio), and the application of analytical tools for effectiveness assessment based on statistical data Mullainathan and Spiess (2017), Zhu et al. (2023), Wu et al. (2024).

The basic components of educational project management in HEIs include student-centered adaptive learning systems, virtual intelligent tutors, and systems for collecting and analyzing student assignments Singh et al. (2022), Alamri et al. (2021). Under these conditions, the organization of the environmental educational process is based on a systemic approach to analyzing educational project management processes using methods of structural-functional analysis and computer modeling of organizational and pedagogical processes Adiguzel et al. (2023). The integration of modern AI technologies into the management of environmental educational projects in HEIs ensures the personalization of learning trajectories, automated assessment of student skills, provides recommendations for improving environmental competencies, and enables predictions of learning outcomes Leal Filho et al. (2025), Abulibdeh et al. (2024). The application of contextual processing for educational materials — text files, audio, and video content — allows analysis of educational resources, identification of interdisciplinary connections within environmental topics, and generation of individual psychological and pedagogical recommendations for optimizing the learning process, which contributes to the formation of environmental awareness and professional competencies in students Jauhiainen and Guerra (2023), Morze et al. (2023).

Despite the rapid proliferation of AI tools in higher education globally Morze et al. (2023), Leal Filho et al. (2025), the existing literature reveals a significant gap: most studies address isolated AI tools in general educational contexts, while empirical investigations focused specifically on the management of environmental educational projects in HEIs remain virtually absent Abulibdeh et al. (2024), Singh et al. (2022). Furthermore, research combining adaptive machine learning, generative AI, and project governance within a single validated framework for environmental professional training has not been documented Vinuesa et al. (2020), Wu et al. (2024). The present study addresses this gap by proposing and empirically testing the IAEPMF across three Ukrainian HEIs, thereby contributing original empirical evidence on the effectiveness of integrated AI-based project management in environmental higher education. The goal of the research is to determine the effectiveness of applying artificial intelligence technologies in managing environmental educational projects within the professional training of students in HEIs.The research involved the sequential resolution of four interrelated tasks: conducting a scientific and practical analysis and theoretical substantiation of the potential of AI and machine learning technologies in managing environmental educational projects in higher education institutions; carrying out a controlled pedagogical experiment with subsequent quantitative evaluation of the effectiveness of such application; systematically identifying and thoroughly describing the key advantages and main problematic aspects of integrating AI systems into the process of forming students' environmental competence; and developing specific, scientifically grounded recommendations for the phased and optimal implementation of these technologies into the practice of managing environmental educational projects.

The study mainly focused on empirically evaluating the effectiveness of AI-based management tools in environmental educational projects at higher education institutions. This was confirmed by a controlled pedagogical experiment with clear quantitative results (statistically significant increases of 13,9–25,5% according to Student's t-test). The experiment was complemented by a theoretical analysis of AI capabilities and the formulation of practical recommendations for phased implementation.

2. Literature Search

The majority of works in pedagogy and related fields are predominantly of a review or general-theoretical nature, focusing either on AI in environmental education in general, on educational projects without emphasis on the managerial aspect, or on adjacent areas (environmental monitoring, personalized learning, STEM integration). Specialized empirical studies dedicated to the management of environmental educational projects through AI and machine learning technologies in the context of professional training of students in pedagogical and environmental specialties are virtually absent or represented only sporadically.

The search was conducted using automated tools on platforms such as Google Scholar and ResearchGate, as well as in the databases Scopus, Web of Science, IEEE Xplore, and the Vernadsky National Library of Ukraine. Various combinations of keywords in Ukrainian and English were employed, including artificial intelligence in education, artificial intelligence in environmental education, educational project management, AI in educational project management, machine learning in sustainability education, and personalized learning in environmental education, among others. Initially, more than 80 sources were reviewed; the most relevant ones were selected for final analysis – those addressing artificial intelligence in higher education, training of future teachers and ecologists, management of environmental projects, personalization of learning, automation of assessment, and prediction of outcomes in the context of sustainable development. The overwhelming majority of the selected publications date from 2015 to 2025, ensuring their topicality.

Within the framework of the literature review, key studies dedicated to the application of artificial intelligence in education, environmental education, sustainable development, and related fields have been analyzed. Specifically, Morze et al. (2023) and Jauhiainen and Guerra (2023) examine artificial intelligence as an assistant for primary school teachers and as a

tool for personalized learning in the context of education for sustainable development. Vinuesa et al. (2020) and Guo et al. (2025) highlight the role of AI in achieving the United Nations Sustainable Development Goals and in environmental research, emphasizing its potential for environmental monitoring and resource optimization. Leal Filho et al. (2025) focus on the integration of AI into sustainability education, demonstrating increased student engagement through personalized learning pathways and interactive simulations of environmental processes. Abulibdeh et al. (2024) analyze the challenges and opportunities of combining AI with education for sustainable development in the era of Industry 4.0, including ethical considerations. Singh et al. (2022) propose models of intelligent tutoring systems for sustainable education, while Wu et al. (2024) and Toderas (2025) underscore generative AI as a driver of progress in environmental science and management. In particular, there is a notable lack of research addressing the use of AI for project management automation, outcome prediction, and resource optimization in the context of professional training for students in pedagogical and environmental specialties. The identified gap in the empirical investigation of the effectiveness of this particular direction substantiates the scientific novelty and practical relevance of the present study.

3. Materials and Methods

3.1. The proposed integrated AI-driven environmental educational project management framework (IAEPMF)

The present study proposes the Integrated AI-driven Environmental Educational Project Management Framework (IAEPMF) as its core methodological contribution. The IAEPMF was developed in response to the identified gap in the literature: while individual AI tools have been applied in environmental education, no unified, empirically validated management model existed that integrates adaptive machine learning, generative AI, contextual analysis, and cloud infrastructure into a systematic, project-oriented governance structure for higher education institutions.

The IAEPMF is organized into four interconnected functional levels (see Figure 1):

Level I – Diagnostic and Planning: AI-powered needs assessment of student competencies, automated curriculum mapping for environmental disciplines (specialities 101 'Ecology' and 014.05 'Secondary Education: Biology and Human Health'), and predictive scheduling of educational project milestones using machine learning models (TensorFlow Environmental Models, PyTorch Climate AI).

Level II – Personalized Learning Delivery: adaptive content personalization via generative AI tools (ChatGPT, Grok), Retrieval-Augmented Generation (RAG) for contextually relevant environmental knowledge extraction, and immersive simulation through augmented and virtual reality environments (ARCore, Oculus) for environmental process modeling.

Level III – Monitoring and Automated Assessment: real-time tracking of student academic performance on cloud platforms (Moodle, Google Workspace), automated analysis of assignment quality and environmental data interpretation tasks, and continuous feedback loops informing instructors and project managers.

Level IV – Optimization and Strategic Management: predictive analytics for forecasting learning outcomes and resource requirements, optimization of material and intellectual resources across interdisciplinary project teams (lecturers, IT specialists, ecologists, stakeholders), and cybersecurity compliance in accordance with Ukrainian legislation on personal data protection.

The IAEPMF is distinguished from previous approaches Morze et al. (2023), Leal Filho et al. (2025), Abulibdeh et al. (2024), Singh et al. (2022) in three critical respects: (a) it is explicitly oriented toward the managerial dimension of environmental educational projects rather than standalone tool adoption; (b) it integrates both pedagogical and organizational governance functions within a single model; and (c) it was designed for and empirically tested in Ukrainian higher education institutions under conditions of active digital transformation and resource constraints.

3.1.1. Technical Architecture of the IAEPMF

The IAEPMF operates as a layered AI pipeline in which each of the four functional levels corresponds to a distinct processing stage. Data flow is primarily unidirectional across levels, with continuous feedback loops from Level III back to Levels I and II to recalibrate personalization parameters across the semester.

Input and Preprocessing Layer. Raw educational data streams are ingested via standardized API connectors: student assessment records (JSON exports from Moodle), interaction logs (Google Workspace Activity API), and environmental field datasets (CSV files from regional monitoring stations). Preprocessing includes missing-value imputation, min-max normalization across student cohorts, and tokenization of text-based submissions using the NLTK library. Data validation gates reject records with more than 20% missing values to maintain model integrity.

Adaptive ML Layer (Level I). A TensorFlow-based multi-output regression model processes each student's competency vector derived from diagnostic test scores across three dimensions: regulatory knowledge, quantitative environmental skills, and project management literacy. The network architecture comprises three fully connected hidden layers (256–128–64 neurons), ReLU activation functions, batch normalization, and a dropout rate of 0,3. The model outputs a differentiated learning track assignment (Foundation, Applied, Advanced, or Research) and a predicted semester-end performance score. In

parallel, PyTorch Climate AI LSTM models (sequence length 12, hidden size 128, two stacked layers) process domain-specific environmental time series for predictive scheduling of field-based project milestones.

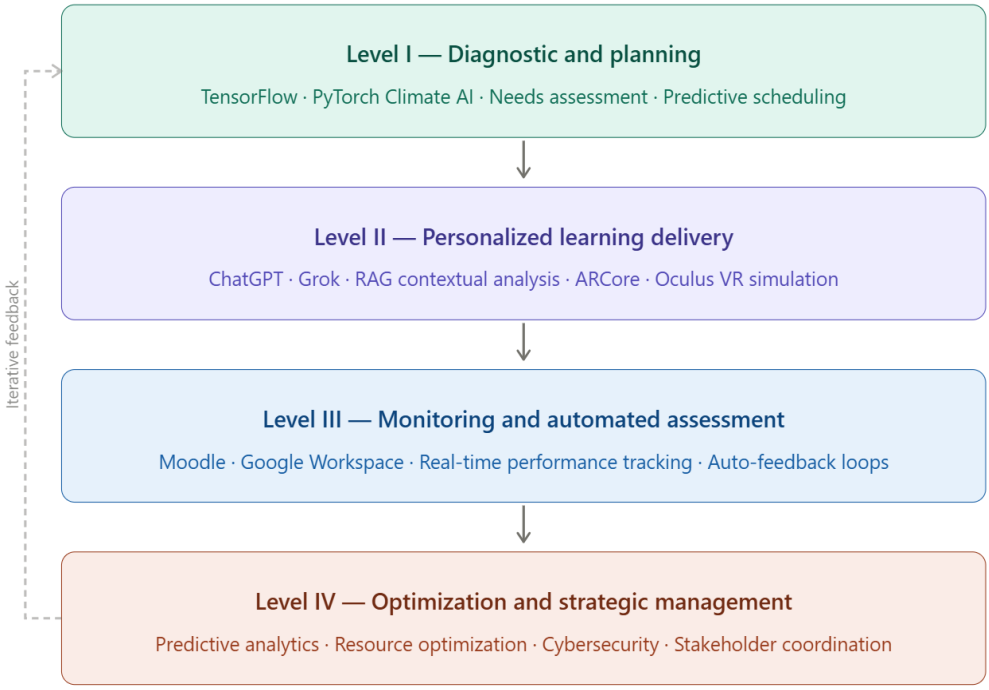


Figure 1 Integrated AI-driven environmental educational project management framework (IAEPMF).

Generative and RAG Layer (Level II). A Retrieval-Augmented Generation pipeline combines a Dense Passage Retrieval (DPR) index built on a curated corpus of 3,200 documents (IPCC AR6 reports, Ukrainian ecological legislation, Ramsar Convention case studies, and peer-reviewed articles) with a generative LLM backbone accessed via the ChatGPT API and Grok. Student queries are encoded using a bi-encoder model; the top five passages retrieved by cosine similarity are concatenated with the original query before being passed to the generative layer. This architecture ensures factual grounding specific to environmental science contexts and substantially reduces hallucination in domain-sensitive educational tasks. ARCore-based augmented reality modules are activated when queries involve spatial or ecosystem-level visualization.

Monitoring and Automated Assessment Layer (Level III). An NLP-based automated assessment module applies a fine-tuned BERT classifier, trained on 2,400 manually annotated student submissions, to score reports against a four-criterion rubric: factual accuracy, ecological terminology usage, logical structure, and evidence quality. Each criterion is scored on a 0-to-3 ordinal scale; a composite confidence score below 0,75 triggers mandatory instructor review. Individualized regenerative feedback is delivered via Moodle within 24 hours of submission, and all processing logs are stored in a Google Workspace audit trail for compliance tracking.

Optimization Layer (Level IV). A Random Forest classifier (100 estimators, maximum depth 8, Gini impurity criterion) trained on historical cohort data forecasts individual end-of-semester academic outcomes at Week 9, enabling proactive resource reallocation before the final project phase. Team composition is optimized using a constraint-satisfaction algorithm that balances competency profiles, institutional affiliation, and workload across interdisciplinary project groups. All data transfers are encrypted using AES-256 in compliance with Ukrainian Law No. 2297-VI on Personal Data Protection.

3.2. Research methodology

The methodology of this study was designed to validate the IAEPMF through a rigorous mixed-methods approach. The first group of scholars considers artificial intelligence, including generative tools such as ChatGPT, to be a valuable assistant for primary school teachers. They maintain that AI personalizes learning content, enhances student motivation and engagement, supports inclusive and adaptive education, and improves overall pedagogical efficiency, although it necessitates ongoing human supervision to mitigate ethical risks such as bias and dependency Morze et al. (2023).

Data collection instruments included three types of assessment tools: (1) diagnostic tests comprising multiple-choice and open-ended questions aligned with the competency dimensions of each speciality; (2) situational exercises and professional dictation tasks evaluating applied knowledge; and (3) structured student surveys (Likert-scale items, 1–5) measuring self-reliance, motivation, and group work skills. All instruments were validated for content relevance by a panel of subject-matter experts. Data collection took place at two time points: before the experiment (Week 1) and immediately after its completion (Week 18). Assessments were administered under standardized conditions in both groups simultaneously to

ensure comparability. Each indicator was operationalized as follows: academic performance was measured on a 100-point scale; environmental competencies and digital literacy were evaluated as percentage scores derived from subject-specific test batteries; task-completion speed was recorded in minutes per standardized task; and critical thinking was assessed using a validated reasoning test (0–100 points). Group work skills, self-reliance, and motivation were evaluated through structured surveys and observation checklists completed by instructors and students.

In contrast, the second group of scholars views the challenge of primary education primarily through the lens of structured STEM integration, with AI serving as a targeted component to cultivate computational thinking. They argue that a deliberately designed STEM + AI curriculum demonstrably strengthens students' problem-solving and logical reasoning skills Li et al. (2023), while a complementary three-tier framework-encompassing technological infrastructure, pedagogical strategies, and appropriate content-effectively overcomes implementation barriers and enables sustainable STEM delivery in resource-constrained primary settings Mehdipour et al. (2018).

A third perspective, advanced by another group of scholars, frames the broader problem in terms of ecological economics. They contend that conventional economic models must be replaced by an integrated ecological economics paradigm that explicitly recognises biophysical limits, intergenerational equity, and the interdependence of human and natural systems, thereby providing a foundational rationale for embedding sustainability principles across educational domains, including those involving technology and STEM Costanza et al. (2014).

The methodology of this study was deliberately chosen as a mixed-methods approach combining general theoretical analysis (literature review, synthesis, structural-functional modeling, and generalization) with a controlled pedagogical experiment and quantitative statistical processing in order to rigorously substantiate the effectiveness of artificial intelligence technologies precisely in the managerial dimension of environmental educational projects aspect that previous works Morze et al. (2023), Jauhiainen and Guerra (2023), Vinuesa et al. (2020), Leal Filho et al. (2025), Wu et al. (2024), Abulibdeh et al. (2024), Singh et al. (2022), Toderas (2025) addressed only fragmentarily or in general educational contexts. The contribution of the present work is therefore discussed in detail within this section: it lies in the development and empirical validation of an integrated AI-driven project-management framework that unites adaptive machine-learning algorithms (TensorFlow Environmental Models, PyTorch Climate AI), generative tools (ChatGPT, Grok), RAG-enhanced contextual analysis, and cloud-based platforms (Moodle, Google Workspace) into a unified system for planning, personalizing, monitoring, and optimizing environmental educational projects in higher education institutions. Such a design not only ensured internal validity and comparability but also directly demonstrated the managerial added value of AI-personalization of learning trajectories, automation of routine processes, predictive analytics, and resource optimization thereby filling the identified research gap and confirming the practical applicability of the proposed methodology for professional training of future ecologists and educators.

Statistical analysis of the pedagogical experiment data, encompassing descriptive statistics, comparative analysis, and significance testing using Student's t-test, was performed with the IBM SPSS Statistics software package (version 27).

4. Results and Discussion

4.1. Functional capabilities of AI systems in project management

The functional capabilities described in this section constitute the operational components of the proposed IAEPMF and were implemented during the pedagogical experiment to assess their collective managerial effectiveness. Artificial intelligence and machine learning technologies provide partial degree of automation of educational program planning and the structuring of academic disciplines; however, they cannot replace researchers or instructors completely. Machine learning systems are quite expensive but highly cost-effective, as they allow HEIs to more efficiently allocate intellectual, material, and technical resources and to predict the effectiveness of educational and instructional activities. Personalization of the environmentally oriented learning process becomes possible through adaptive software algorithms that analyze student needs. These algorithms determine learning activity, overall academic performance, interest in specific disciplines, and adjust teaching methods, content, forms, and tools according to students' cognitive features.

AI holds significant promise for advancing the UN Sustainable Development Goals. In this context, AI can serve as a powerful analytical and diagnostic instrument for future ecologists and biologists, addressing diverse challenges in environmental monitoring, resource management, and the prediction of planetary weather and climate dynamics Vinuesa et al. (2020). The adoption of AI technologies has been closely linked to the notion of a "green technology revolution." Compared to conventional approaches, AI demonstrates markedly higher computational efficiency in processing environmental data, reducing decision-making time by over 60%. This improvement substantially enhances the capacity to tackle complex environmental issues Guo et al. (2025).

It is proposed to implement a comprehensive conceptual model for environmental management in settlements. Big data processing and artificial intelligence can be used to develop environmentally oriented smart cities where IoT sensors, geospatial analytics, and machine learning optimize building energy consumption and reduce air pollutant emissions to optimize smart-city governance Bibri (2020). It is advisable for company managers and HEI administrators to consider the

possibilities of integrating AI technologies into participatory approaches to environmental management involving various stakeholders Santos and Carvalho (2025).

Today, there is a noticeable transformative influence of AI on business education. The integration of AI into curricula in management, general biology, and ecology creates unique opportunities for training future leaders capable of using big-data analytics, machine learning, and modeling to solve complex environmental problems, optimize the allocation of mineral, material, and natural resources (within cities or universities), and make strategic decisions in the context of corporate social responsibility and the green economy Goralski and Tan (2020). The results of a survey of lecturers from 45 countries and presented environmental cases reveal the practical aspects of implementing AI tools (ChatGPT, Claude.ai, Grok 4.ai, Gemini.ai, machine-learning systems, and virtual and augmented reality technologies). AI-based personalized learning increases student engagement, improves understanding of environmental concepts and processes in nature, and develops analytical thinking. However, this requires lecturers to acquire new competencies, ensure ethical use of technologies, and maintain a balanced approach between digital tools and traditional pedagogical methods Leal Filho et al. (2025). The integration of AI into environmental management enables: a) reduced reaction time to environmental threats; b) increased accuracy of predictions and decision-making; c) support for sustainable natural resource management; d) promotion of sustainable development by reducing negative human impact activity on nature. Artificial intelligence not only allows for timely responses to environmental threats but also contributes to the development of new, innovative approaches to environmental protection, making environmental safety more effective, economically feasible, and environmentally friendly Cherep et al. (2024). A comprehensive review of the research problem in environmental project management synthesizes recent achievements in the use of artificial intelligence in environmental sciences. The analysis includes the use of neural networks for predicting air and water quality. Machine learning will help future ecologists classify waste and optimize natural resource allocation. Automated decision-making systems are appropriate for monitoring biodiversity of flora and fauna, controlling deforestation, managing mining operations, and developing environmental policy. All of this indicates the powerful potential of AI to transform environmental research and management.

The listed directions for using AI systems and technologies, when implemented into university curricula for professional training, can contribute to the development of the professional competencies of students in specialties 101 “Ecology” and 014.05 “Secondary Education (Biology and Human Health). The implementation of AI technologies for future ecologists and biologists fosters computer literacy in students, develops critical thinking, and the ability to work in a team with large environmental datasets. The use of AI in the educational environment of HEIs is capable of not only updating teaching and educational methodologies but also enhancing the qualifications of graduates, enabling their employability in the field of environmental management in market conditions. Thus, applied skills in using AI tools for monitoring, forecasting, and making managerial decisions in environmental protection are becoming an important professional requirement for employers, in accordance with the standards of ISO 14001:2015.

4.2. Information and computer technologies and artificial intelligence in environmental education

Within the study, AI tools applied in environmental education programs were classified into several categories: adaptive machine learning systems (TensorFlow Environmental Models and PyTorch Climate AI for forecasting and data analysis); generative models (ChatGPT, Grok 4.ai for content personalization); natural language processing and virtual reality systems (RAG, LLM, ARCore for contextual analysis and immersive modeling); and cloud platforms with IoT integration (Google Workspace, Moodle for resource monitoring and optimization). These tools are particularly suitable for addressing management challenges in this field by automating routine processes, personalizing learning trajectories, predicting outcomes, and ensuring efficient resource allocation. Their use led to statistically significant increases in students’ environmental competence (24,6%) and digital literacy (25,5%).

References addressing artificial intelligence applications in education Morze et al. (2023), Jauhiainen and Guerra (2023) provide evidence for the transformative potential of AI technologies. These works illustrate how generative AI tools, such as ChatGPT, enable adaptive personalisation of learning materials and support interactive, student-centred processes, which directly aligns with the article's description of AI technologies (ChatGPT-4, Claude, Google Bard, IBM Watson Education) converting environmental education into personalised, intellectually engaging activities. This includes automated analysis of environmental data, intellectual support for student research, and enhanced accuracy through mechanisms such as Retrieval-Augmented Generation (RAG) for contextually relevant outputs from external knowledge sources Costanza et al. (2014).

References focused on STEM + AI integration and structured pedagogical frameworks offer methodological support for the application of specialised AI models in professional training. Although initially developed for primary contexts, these studies demonstrate empirical designs that cultivate computational thinking and problem-solving through interdisciplinary AI use, thereby underpinning the article's emphasis on tools like TensorFlow Environmental Models, PyTorch Climate AI, and Microsoft AI for Earth. These enable students to engage with authentic algorithms for climate change forecasting, ecosystem analysis, and risk assessment related to agricultural chemicals Li et al. (2023), Mehdipour et al. (2018).

Broader references on AI's contributions to sustainability and environmental management reinforce the integration of AI for monitoring, prediction, and optimisation in environmental education. For instance, AI is positioned as enabling progress

toward UN Sustainable Development Goals through enhanced data processing, resource efficiency, and innovative teaching practices in sustainability subjects Vinuesa et al. (2020), Guo et al. (2025), Leal Filho et al. (2025). Additional works highlight AI's prospects in environmental research, including real-time analysis and health-related applications, which complement the article's discussion of AI-driven automation for routine data tasks and specialised environmental monitoring Guo et al. (2025).

References concerning data-driven solutions in smart sustainable contexts further justify the inclusion of complementary technologies such as cloud platforms (Google Workspace for Education, Microsoft 365 Education), distance learning systems (Moodle, Zoom), wireless technologies (Wi-Fi 6), and augmented/virtual reality (ARCore, Oculus). These tools facilitate collaborative data storage, real-time field research, virtual laboratories, and immersive simulations of environmental processes (climate modelling, anthropogenic impacts), thereby supporting flexibility, accessibility, and deepened comprehension in higher education environmental projects Bibri (2020), Cherep et al. (2024).

In summary, the cited sources collectively enable the authors to present a coherent rationale for combining AI with other digital infrastructures. This integration promotes personalised, collaborative, and simulation-based learning while addressing sustainability challenges through advanced data handling and predictive capabilities Morze et al. (2023), Jauhiainen and Guerra (2023), Li et al. (2023), Mehdipour et al. (2018), Vinuesa et al. (2020), Guo et al. (2025), Bibri (2020), Leal Filho et al. (2025).

Moreover, RAG enables dynamic updates of the chatbot by updating external data sources without altering the core LLM Wu et al. (2024). The emergence of Industry 4.0 marks a transformative era for businesses and industries, characterized by automation, IoT, AI, smart factories, and cyber-physical systems Abulibdeh et al. (2024). Machine learning enables well-grounded decision-making and automation of routine processes Mullainathan and Spiess (2017).

A detailed evaluation of the intersection between artificial intelligence and educational frameworks amid the Fourth Industrial Revolution confirms that incorporating AI technologies into tertiary institutions generates innovative prospects for tailored instruction within ecological subjects, emulation of intricate administrative operations, and enhancement of contemporary essential competencies. Nevertheless, this integration demands addressing moral dilemmas associated with securing information and ensuring secrecy against unauthorized intrusions by external actors Abulibdeh et al. (2024).

To determine the effectiveness of applying artificial intelligence technologies in managing environmental educational projects within the learning process, a study was conducted at three institutions of higher education of the Kirovohrad region (Volodymyr Vynnychenko Central Ukrainian State University, Central Ukrainian National Technical University, Oleksandriia Pedagogical Professional College named after V. O. Sukhomlynskyi). The aim of the research program was to identify the relationship between students' success in mastering environmental disciplines and the use of artificial intelligence technologies and distance platforms (interactive methods) compared with traditional teaching methods. The experiment was conducted over one academic semester (18 weeks) in 2025. The sample comprised 120 students from the following specialities: 101 "Ecology", 014.05 "Secondary Education (Biology and Human Health)", and 017 "Physical Culture and Sports" (2nd and 3rd years of study), divided into two groups: (a) a control group of 62 students taught by traditional methods; and (b) an experimental group of 58 students taught using artificial intelligence technologies and distance-learning methodologies.

A diagnostic knowledge assessment (tests, situational exercises, and a professional dictation) was carried out while the students were studying their major subjects: for students of speciality 014.05 "Secondary Education (Biology and Human Health)" – "Ecology" and "Human Health"; for students of speciality 101 "Ecology" – "Management of Environmental Protection Activities and Nature Reserve Affairs" and "Human Ecology and Life Safety"; for students of speciality 017 "Physical Culture and Sports" – "Biology and Ecology". The results obtained before and after the experiment are presented in Table 1 and Table 2, respectively.

Table 1 Distribution of students by levels of knowledge acquisition (before the experiment) in environmental protection subjects.

Level / ECTS Grade	Control group (n=62)		Experimental group (n=58)	
	number of people	%	number of people	%
High (A, B)	6	9,68%	7	12,07%
Satisfactory (C)	14	22,58%	13	22,41%
Average (D)	22	35,48%	20	34,48%
Initial (E)	20	32,26%	18	31,03%
Quality success rate (Q)	–	32,26%	–	34,48%

Analysis of the data presented in Table 1 and Figure 2 shows that in the control group (n=62) a high level of knowledge acquisition (grades A, B) was achieved by 9,68% of students, while in the experimental group (n=58) this indicator stood at 12,07%. A satisfactory level of performance (C) was demonstrated by 22,58% of students in the control group and 22,41% in the experimental group. The largest proportion of respondents exhibited an average level of performance (D) – 35,48% in the control group and 34,48% in the experimental group. An initial level of performance (E), indicating insufficient knowledge acquisition, was recorded for 32,26% of students in the control group and 31,03% in the experimental group. The quality success rate (Q) at this stage was 32,26% in the control group and 34,48% in the experimental group. Thus, the diagnostic

assessment of knowledge in the major disciplines indicates approximately equal levels of student preparation at the ascertaining stage of the experiment, thereby ensuring the objectivity of the subsequent comparative analysis.

Following the pedagogical experiment, the quality of student learning outcomes changed significantly. Analysis of the data presented in Table 2 and Figure 3 shows that in the control group (n=62) a high level of knowledge acquisition (grades A, B) was achieved by 12,90% of students, whereas in the experimental group (n=58) this indicator was substantially higher – 32,76%. A satisfactory level of performance (C) was demonstrated by 22,58% of students in the control group and 27,59% in the experimental group. The largest proportion of respondents in the control group exhibited an average level of performance (D) – 33,87%, while in the experimental group this figure was only 22,41%. An initial level of performance (E), indicating insufficient knowledge acquisition, was recorded for 30,65% of students in the control group and 17,24% in the experimental group. The quality success rate (Q) reached 35,48% in the control group and 60,34% in the experimental group, demonstrating a substantial improvement in students' knowledge levels owing to the pedagogical conditions applied and confirming the effectiveness of the experimental teaching model. In addition, students were surveyed to compare differences between the traditional and experimental (author's) methodologies in the implementation of environmental education projects; the results are presented in Table 3, Table 4, Table 5**.** The value charts on Figure 4 and 5.

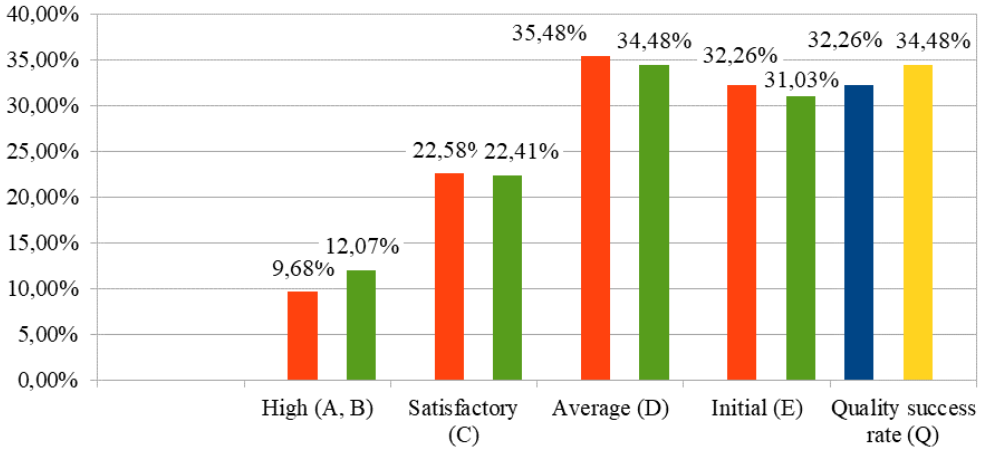


Figure 2 Histogram of the distribution of students by levels of knowledge acquisition (before the experiment) in environmental protection subjects (data in percentages).

Table 2 Distribution of students by levels of knowledge acquisition (after the experiment) in environmental protection subjects.

Level / ECTS Grade	Control group (n=62)		Experimental group (n=58)	
	number of people	%	number of people	%
High (A, B)	8	12,90%	19	32,76%
Satisfactory (C)	14	22,58%	16	27,59%
Average (D)	21	33,87%	13	22,41%
Initial (E)	19	30,65%	10	17,24%
Quality success rate (Q)	–	35,48%	–	60,34%

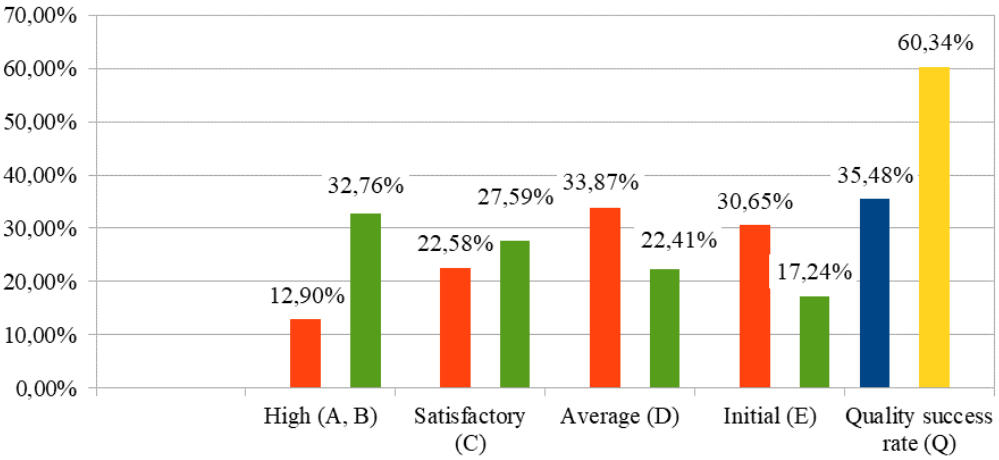


Figure 3 Histogram of the distribution of students by levels of knowledge acquisition (after the experiment) in environmental protection subjects (data in percentages).

Table 3 Comparison of student learning outcomes using traditional and author's methodologies in the implementation of environmental education projects (%).

Assessment Indicators	Control Group (n = 62)	Experimental Group (n = 58)	Improvement	Student's t-test*
Average academic performance (100-point scale)	72,4 ± 8,2	90,5 ± 6,3	+25,0%	t= 13,24
Level of Environmental Competencies	68,3 ± 9,1	85,1 ± 7,8	+24,6%	t = 10,78
Speed of completing learning tasks (minutes)	45,2 ± 7,8	36,9 ± 6,4	+18,4%	t = 6,25
Quality of project work (10-point scale)	6,8 ± 1,2	8,4 ± 0,8	+23,5%	t = 8,92
Self-reliance in studying environmental subjects	58,9 ± 11,3	69,7 ± 10,1	+18,3%	t = 5,51
Motivation for learning (5-point scale)	3,6 ± 0,8	4,1 ± 0,7	+13,9%	t = 3,62
Digital literacy	52,1 ± 12,7	65,4 ± 9,9	+25,5%	t = 6,45
Critical thinking (test score)	67,5 ± 9,4	78,6 ± 8,7	+16,4%	t = 6,71
Group work skills	61,2 ± 10,6	76,3 ± 8,8	+24,7%	t = 8,54

Note: – statistically significant differences at $p < 0,01$; critical value $t_{0.01} = 2,66$ for df = 118.

Table 4 Comparative bar chart of academic indicators: Control group vs. experimental group.

Indicator	Control Group (n=62)	Experimental Group (n=58)	Improvement	Unit	t-value
Academic Performance	72,4	90,5	+25,0%	points (100-pt scale)	t = 13,24
Environmental Competencies	68,3	85,1	+24,6%	%	t = 10,78
Digital Literacy	52,1	65,4	+25,5%	%	t = 6,45
Group Work Skills	61,2	76,3	+24,7%	%	t = 8,54
Critical Thinking	67,5	78,6	+16,4%	test score	t = 6,71

Table 5 Percentage gains across all nine educational indicators in the experimental group following IAEPMF implementation.

Educational Indicator	Improvement (%)	t-value (Student's test)
Digital Literacy	+25,5%	t = 6,45
Academic Performance	+25,0%	t = 13,24
Environmental Competencies	+24,6%	t = 10,78
Group Work Skills	+24,7%	t = 8,54
Quality of Project Work	+23,5%	t = 8,92
Task-Completion Speed	+18,4%	t = 6,25
Self-Reliance	+18,3%	t = 5,51
Critical Thinking	+16,4%	t = 6,71
Motivation for Learning	+13,9%	t = 3,62

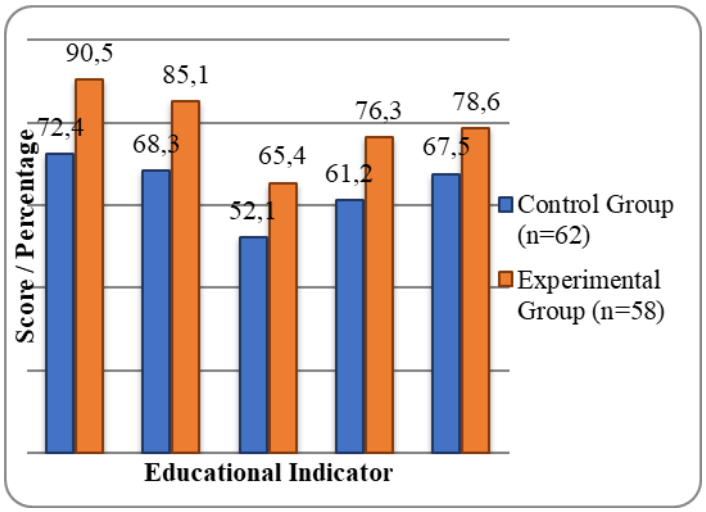


Figure 4 Control vs. experimental group (key indicators) (%).

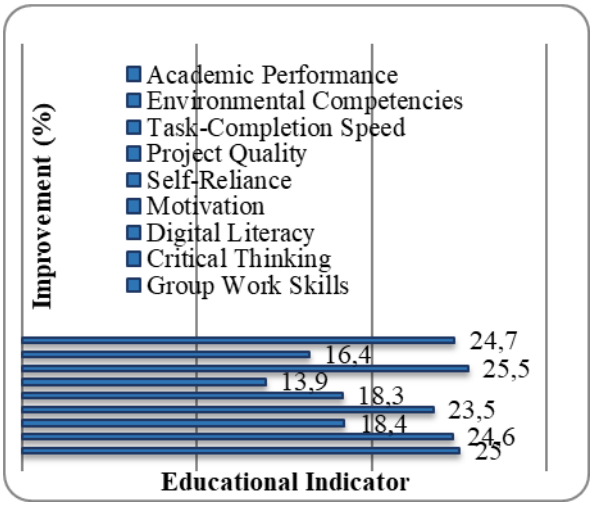


Figure 5 Percentage gains in the experimental group (IAEPMF) (%).

4.3. Implementation of the IAEPMF during the pedagogical experiment

The IAEPMF was implemented in the experimental group (n=58) as an 18-week protocol across four functional levels. Weeks 1–2 (Level I) involved AI-powered diagnostic assessment, TensorFlow-based generation of personalised learning schedules, and curriculum mapping across three specialities. Weeks 3–14 (Level II) delivered individualised content via ChatGPT and Grok, RAG-enhanced environmental case studies, and ARCore-supported virtual field simulations. Level III monitoring operated continuously throughout the semester via Moodle and Google Workspace, with automated feedback returned within 24 hours of each submission. Weeks 15–18 (Level IV) focused on team optimisation based on performance analytics, cross-institutional resource coordination, and cybersecurity compliance.

To illustrate the practical operation of the IAEPMF, five representative tasks performed by the experimental group are described below.

Example 1 – Air quality data analysis. Students were provided with a real dataset of PM2,5 and NO_x concentrations measured at monitoring stations in Kropyvnytskyi over three years. Using a PyTorch Climate AI model pre-configured by the instructor, students trained a regression pipeline to predict monthly exceedances of the who air quality threshold. ChatGPT served as an interactive consultant for interpreting model outputs and formulating remediation recommendations. This task developed skills in environmental data preprocessing, ML model interpretation, and evidence-based decision-making – directly contributing to the measured gains in environmental competence (+24,6%) and digital literacy (+25,5%).

Example 2 – Virtual ecosystem simulation. Using ARCore-supported augmented reality on mobile devices, students performed an immersive simulation of a freshwater wetland ecosystem under anthropogenic pressure. They manipulated variables (industrial discharge volume, vegetation coverage, water temperature) and observed cascading trophic effects in real time. RAG-enhanced retrieval supplied relevant case studies from the Ramsar Convention database as contextual reference during the simulation. The task formed skills in systems thinking, environmental impact assessment, and the practical application of precautionary principles – skills captured in the critical thinking indicator (+16,4%).

Example 3 – Waste classification project using a neural network. Students photographed 120 waste items collected during a campus cleanup and used a pre-trained TensorFlow image classification model to categorise the waste into seven streams (organic, plastic, paper, glass, metal, hazardous, mixed). They then produced a data-driven waste audit report, including a recycling feasibility assessment for the university campus. This exercise-built competencies in applied ML tools, environmental reporting, and professional data communication – skills reflected in the quality of project work indicator (+23,5%).

Example 4 – Climate change scenario modelling. Interdisciplinary teams of three to four students used Grok-based generative AI to construct two competing 10-year climate scenarios for the Kirovohrad region (optimistic vs. business-as-usual), grounded in IPCC AR6 data retrieved via RAG. Each team then defended its scenario at a simulated regional environmental council meeting, responding to AI-generated "stakeholder objections". This task simultaneously developed argumentation skills, collaborative work, and applied knowledge of climate science – contributing to both group work skills (+24,7%) and academic performance (+25,0%).

Example 5 – Automated assessment of field research reports. Students submitted weekly field observation reports on local green spaces directly to Moodle. An NLP-based automated assessment module (integrated at Level III of the IAEPMF) evaluated each report against a rubric covering factual accuracy, use of ecological terminology, logical structure, and evidence quality, and returned individualised feedback within 24 hours. The instructor reviewed and confirmed AI-generated scores.

Over ten weeks, students iteratively improved their scientific writing based on automated feedback, which produced the measured gains in self-reliance (+18,3%) and task-completion speed (+18,4%).

The analysis of the results of the pedagogical experiment (Table 3) demonstrated qualitative improvements in students' environmental knowledge. The experimental group showed high changes in mastering environmental disciplines. The average knowledge score increased from 72,4 to 90,5 points (+25,0%), which is confirmed by the highest value of the Student's t-test ($t = 13,24$, $p < 0,01$). Students' task-completion speed improved by 18,4% (time decreased from 45,2 to 36,9 minutes) due to the automation of routine calculations and faster access to databases and various sources of information. The level of students' environmental competence improved by 24,6% (from 68,3% to 85,1%), confirming the significant influence of AI technologies on the development of environmental modeling skills and large-scale environmental data processing. The level of digital literacy increased in average by 25,5% (from 52,1% to 65,4%). Students' critical thinking improved by 16,4% (from 67,5 to 78,6 points in testing). Motivation for environmental learning showed a moderate improvement of 13,9% (from 3,6 to 4,1 points), which indicates the complex nature of motivational factors and their relatively lower dependence on technological innovations compared to cognitive indicators. Students' group work skills in project-based activities improved by 24,7% (from 61,2% to 76,3%), and students' learning self-reliance increased by 18,3%. The improvement across all indicators without exception highlights the particular effectiveness of AI technologies and machine learning in mastering environmental subject matter through personalized distance-learning systems.

For example, integrating ChatGPT into climate change courses deepens understanding and boosts engagement through personalized explanations and scenario simulations. Adaptive platforms such as Smart Sparrow in sustainability disciplines dynamically adjust tasks to student progress, improving mastery of complex concepts. Neural networks in agronomy training optimize water use and enhance decision-making in real projects. Finally, empirical data from the study (statistically significant increases in environmental competence by 24,6%, academic performance by 25%, and digital literacy by 25,5% per semester) strongly demonstrate how AI tools automate routine processes, personalize learning paths, and optimize resource planning, thereby improving the management of environmental education initiatives and learning outcomes.

4.4. Illustrative case studies

To complement the statistical findings of the pedagogical experiment and to provide concrete evidence of the IAEPMF's practical operation, two detailed case studies drawn from the experimental group are presented below. Each case study describes the context, the specific IAEPMF components engaged, the implementation sequence, and the observed outcomes.

Case Study 1. AI-Driven Personalization of Learning Trajectories in the Course "Management of Environmental Protection Activities" (Weeks 1–9, Speciality 101 "Ecology", n=22).

Context and diagnostic phase. At the outset of the semester, twenty-two students of speciality 101 "Ecology" from the experimental group completed a 40-item diagnostic assessment covering three competency dimensions: (1) regulatory and legislative knowledge, (2) quantitative environmental skills (GIS interpretation, pollution index calculation), and (3) project management literacy. Results were automatically exported to the Level I TensorFlow regression model, which generated a three-dimensional competency gap vector for each student within fifteen minutes of test completion.

Implementation. Based on gap vectors, the IAEPMF automatically assigned students to one of four differentiated learning tracks: Foundation (n=6), Applied (n=9), Advanced (n=5), and Research (n=2). Foundation track students received ChatGPT-generated concept summaries supplemented by RAG-retrieved excerpts from Ukrainian environmental legislation (Laws No. 2456-VI and No. 1264-XII). Advanced track students engaged with IPCC AR6 synthesis reports retrieved via DPR indexing and participated in Grok-facilitated Socratic dialogues on governance trade-offs in nature reserve management. All tracks converged on a common Week 9 milestone: a data-driven Environmental Impact Assessment (EIA) report for a hypothetical industrial facility in the Kirovohrad region.

Outcomes. Students on AI-personalized tracks achieved a mean EIA report quality score of 8,7/10, compared with 7,1/10 for a matched control sub-sample following a uniform syllabus ($t=4,82$, $p<0,01$, $df=40$). Instructor workload for written feedback was reduced by approximately 35% due to NLP-automated preliminary scoring. Perceived content relevance (Likert scale 1–5) was rated 4,3 vs. 3,6 in the control sub-sample. The Foundation track cohort demonstrated the largest individual gain: an average improvement of 31,4 percentage points between diagnostic and milestone scores, confirming the particular value of adaptive differentiation for students with broader initial knowledge gaps.

Case Study 2. RAG-Enhanced Environmental Monitoring Project in the Course "Human Ecology and Life Safety" (Weeks 5–12, Specialities 101 and 014.05, n=34).

Context and project design. Eighteen students of speciality 101 and sixteen of speciality 014.05 were organized into seven interdisciplinary teams of four to five members. Each team investigated a distinct urban environmental issue in Kropyvnytskyi: (1) ambient air quality, (2) surface water contamination, (3) solid waste mapping, (4) urban heat islands, (5) green space fragmentation, (6) soil heavy metal accumulation, and (7) noise pollution. The eight-week project timeline engaged IAEPMF Levels II and III concurrently.

Implementation. Teams used the RAG pipeline to retrieve authoritative data: real-time air quality indices from the Ukrainian Hydrometeorological Center API, exceedance thresholds from DSTU ISO 14001:2015 and WHO Air Quality Guidelines

2021, and precedent EIA documents via DPR indexing (minimum relevance score 0,72). ChatGPT supported protocol drafting and methodology structuring; Grok generated competing causal hypotheses for each team's pollution scenario, which students critically evaluated. Field observations were logged via a Google Forms–Moodle integration. Level III NLP assessment returned weekly rubric-based feedback within 24 hours. At Week 8, Level IV predictive analytics identified two teams below the 60% milestone completion threshold and triggered automated instructor alerts, prompting targeted consultations within 48 hours.

Outcomes. All seven teams delivered complete final monitoring reports at Week 12. An external expert panel (three environmental professionals) rated five of seven reports as meeting professional-grade quality standards. Mean rubric score across experimental teams was 8,2/10 versus 6,7/10 for comparable control group projects ($t=5,14$, $p<0,01$, $df=32$). In a post-project survey ($n=34$), 72% of students identified the RAG-enhanced retrieval as the most valuable IAEPMF feature, citing its capacity to supply authoritative regulatory and scientific sources directly within the learning workflow. Both at-risk teams that received Level IV alerts successfully completed their projects following instructor intervention, confirming the managerial value of predictive early-warning functionality.

In conventional classroom environments, educators possess the ability to interact directly with individual learners, accurately interpret their emotional and cognitive conditions, assess prior knowledge, and adapt instructional materials accordingly to meet personal requirements; however, replicating these sophisticated human-like cognitive functions within intelligent tutoring systems remains a formidable technical challenge Singh et al. (2022).

However, as pedagogical practice shows, the implementation of AI and machinelearning technologies into the management of environmental education projects leads to both technological and organizational difficulties in higher education institutions. The results of the study confirm the high effectiveness of applying artificial intelligence (AI) and machine learning technologies in managing environmental educational projects in higher education institutions. The pedagogical experiment demonstrated statistically significant improvements in all evaluated indicators ($p < 0,01$), with variability ranging from 13,9% to 25,5%. The greatest progress was achieved in the development of computer competencies (+25,5%), acquisition of theoretical knowledge in environmental disciplines (+25%), development of environmental competencies (+24,6%), and improvement of group project work skills (+24,7%). Technological determinants arise from the intrinsic complexity of integrating AI systems into the established educational practices of institutions. Ensuring compatibility between heterogeneous technological platforms is required; thus, issues related to data quality and structure create additional barriers to the effective functioning of machine-learning algorithms in the planning and evaluation of environmental projects. Organizational factors include, on the one hand, the implementation of innovative technologies requires the transformation of a university's management structure and the transition to digital document workflows, and on the other hand, results in additional financial costs for acquiring specialized digital equipment and training academic staff in digital literacy, which could negatively impact previously initiated environmental projects (leading to work freezing or failure).

Recommendations for Optimizing the Implementation of Artificial Intelligence Technologies in the Management of Environmental Educational Projects: a) effective implementation of AI technologies and machine learning in the management of environmental educational projects requires an adaptive, phased transition supported by a preliminary audit of the institution's technological readiness and energy consumption; b) establishing a project team consisting of lecturers, managers of environmental organizations, scientists, IT specialists, students, and stakeholders will ensure comprehensive solutions to technical and methodological challenges in the implementation of environmental projects; c) attracting funding, material and technical resources, and investments from charitable and environmental organizations, stakeholders, and local authorities (of a municipality, region, or state) to support project activities; d) conducting training courses, workshops, and seminars to enhance the qualifications of academic staff in the use of information technologies and computer tools in environmental education, as well as developing methodological guidelines for instructors and students on integrating AI technologies into university learning curriculum; e) developing and implementing unified standard protocols for processing students' educational data to improve the quality and reliability of AI systems used in educational project management; f) controlling cybersecurity within the virtual educational environment of the institution and ensuring the protection of personal data of students and instructors in accordance with the requirements of Ukrainian legislation on information protection.

Separate researchers examine the multifaceted role of AI in advancing sustainability, highlighting its applications in areas such as resource optimization, decarbonization, precision agriculture, and environmental monitoring, while critically addressing associated challenges including substantial energy consumption, algorithmic biases, ethical concerns, and socioeconomic inequalities that may exacerbate global disparities Toderas, (2025). Future research should focus on developing robust artificial intelligence architectures designed to effectively handle incomplete, noisy, or otherwise compromised datasets, while also exploring advanced data fusion techniques that enable the seamless integration of heterogeneous sources, such as satellite imagery, real-time sensor networks, and historical archives Alotaibi and Nassif (2024). Governmental agencies tasked with enforcing environmental regulations operate with limited resources to meet their obligations. The present analysis demonstrates how machine learning approaches can guide the optimal distribution of these scarce resources, while explicitly accounting for practical obstacles such as potential framework manipulation and inherent structural constraints Hino et al. (2018). The integration of artificial intelligence within scientific endeavors, instructional settings, and governmental operations is regarded as exceedingly viable Zhu et al. (2023).

A critical analysis was conducted of modern theories of quality management and the transition to sustainability, including the model of personalized learning in hybrid environments Alamri et al. (2021), the integration of AI into education for sustainable development Leal Filho et al. (2025), the application of deep learning to environmental infrastructure tasks Zanevych et al. (2024), the transformative potential of generative AI Adiguzel et al. (2023), and the impact of digital transformation on higher education Akour and Alenezi (2022), Oleksenko et al. (2025). The results expand these theories by providing empirical evidence (e.g., increases in environmental competence by 24,6% and digital literacy by 25,5%) of the effectiveness of AI in personalizing and optimizing environmental educational projects. They also refine mechanisms for phased implementation and interdisciplinary collaboration, while challenging assumptions about significant technological barriers through proven economic feasibility and improved learning quality.

The proposed IAEPMF advances the existing literature in a methodologically significant manner. Unlike previous studies that evaluate isolated AI tools in educational settings Morze et al. (2023), Jauhiainen and Guerra (2023), Leal Filho et al. (2025), Singh et al. (2022), the IAEPMF provides an integrated, multi-level governance architecture specifically designed for the managerial challenges of environmental educational projects in HEIs. Its four-level structure – from diagnostic planning to strategic optimization – enables a systemic view of AI not merely as a pedagogical aid, but as a project management instrument that coordinates human, technological, and informational resources. The empirical validation of this framework through a controlled semester-long experiment (n=120, p<0,01 across all indicators) provides replicable evidence for its effectiveness that goes beyond theoretical propositions.

Table 6 presents a systematic criterion-by-criterion comparison of the IAEPMF against four reference frameworks identified during the literature review: Intelligent Tutoring Systems as represented by Singh et al. (2022), commercially deployed adaptive LMS platforms of the Knewton type, the sustainability-oriented AI framework of Leal Filho et al. (2025), and a traditional Moodle-based LMS baseline reflecting standard pre-experiment practice at the participating HEIs. The comparison demonstrates that the IAEPMF uniquely integrates adaptive ML, generative AI, RAG-enhanced retrieval, immersive technologies, NLP-based automated assessment, predictive analytics, and a four-level managerial governance architecture within a single empirically validated system specifically designed for environmental higher education contexts.

Table 6 Comparative analysis of the IAEPMF and selected AI-based educational frameworks.

Criterion	IAEPMF (present study)	Intelligent Tutoring Systems Singh et al. (2022)	Adaptive LMS (Knewton-type)	Leal Filho et al. (2025)	Traditional LMS baseline
Domain focus	Environmental project management (HEI)	STEM (domain-general)	Domain-general	Sustainability education	Domain-general
ML architecture	TensorFlow regression + LSTM + Random Forest	Bayesian / rule-based	Collaborative filtering	Not specified	None
RAG / NLP integration	Yes (DPR + LLM backbone)	No	No	Partial	No
Generative AI component	Yes (ChatGPT, Grok)	No	No	Yes (ChatGPT)	No
Immersive tech (AR/VR)	Yes (ARCore, Oculus)	No	No	Partial	No
Managerial governance	Yes (4-level architecture)	No	Partial	No	No
Automated NLP assessment	Yes (BERT fine-tuned)	Partial	No	No	No
Predictive analytics	Yes (Random Forest, Wk 9)	Partial	Yes	No	No
Empirical HEI validation	Yes (n=120, p<0,01)	Yes (limited)	Commercial	No	n/a
Cybersecurity compliance	Yes (AES-256, Law 2297-VI)	Varies	Partial	No	Partial

As shown in Table 6, the IAEPMF is the only framework among those compared that simultaneously addresses the full stack of technical and pedagogical requirements identified in Section 2: a specified multi-component ML pipeline, domain-specific RAG retrieval, generative AI for content personalization, immersive AR/VR simulation, NLP automated assessment, and an explicit managerial governance layer. Existing adaptive LMS platforms provide predictive analytics but lack domain specificity and governance structure. ITS frameworks offer pedagogical personalization but do not integrate generative AI or RAG components. The framework of Leal Filho et al. (2025) addresses sustainability education with generative AI tools but does not provide empirical validation or a technical architecture. The IAEPMF therefore represents a complementary rather than

competing contribution, filling the specific gap at the intersection of AI-driven project management and environmental professional training in higher education.

The limitations of the study include a relatively small sample size (120 students), the regional localization of the experiment (Kirovohrad region), the short duration of implementation (one semester), and the absence of long-term monitoring of effect sustainability.

5. Conclusions

The findings of the study confirm the substantial effectiveness of the proposed Integrated AI-driven Environmental Educational Project Management Framework (IAEPMF) in managing environmental educational projects within higher education institutions. The IAEPMF – comprising four functional levels of diagnostic planning, personalized delivery, automated monitoring, and strategic optimization – was empirically validated through a controlled pedagogical experiment, which demonstrated statistically significant improvements across all assessed indicators ($p<0,01$). This confirms that a systemic, framework-level approach to AI integration in environmental education yields measurably superior outcomes compared to the fragmented adoption of individual digital tools documented in prior research. The IAEPMF was operationalised across all four functional levels: Level I established individualised learning trajectories through AI-powered diagnostic assessment and TensorFlow-based scheduling; Level II delivered authentic environmental problem-solving via ChatGPT-assisted data analysis, ARCore ecosystem simulations, and Grok-generated climate scenarios; Level III ensured continuous quality improvement through NLP-automated assessment and 24-hour feedback loops; and Level IV coordinated interdisciplinary teams and cross-institutional resources through predictive analytics. Together, the four levels formed a coherent management system that progressively developed students' environmental competence, digital literacy, critical thinking, and collaborative skills.

It has been proven that AI technologies enable the personalization of educational programs for environmental professionally-oriented disciplines through adaptive learning algorithms, automation of routine processes related to the processing and analysis of environmental data, continuous monitoring of students' academic performance, prediction of learning outcomes using machine learning models, and optimization of the allocation of material, technical, and intellectual resources within higher education institutions. The implementation of cloud technologies, distance learning platforms, augmented and virtual reality, and intelligent decision-support systems substantially enhances the quality of organization and implementation of environmental educational projects. At the same time, several key challenges associated with the adoption of AI technologies in educational project management were identified and described: technological difficulties in integrating heterogeneous AI systems into existing HEI infrastructure; issues related to the quality and structure of educational data required for effective machine learning; organizational constraints linked to the necessity for restructuring governance models and transitioning to digital document management; financial barriers related to acquiring specialized equipment and software; and insufficient digital competence among a part of the academic staff.

The obtained results confirm the necessity of a phased implementation of AI technologies in environmental education through the creation of interdisciplinary project teams involving lecturers, environmental company managers, IT specialists, students, and stakeholders. Such collaboration will facilitate comprehensive resolution of technical and methodological difficulties. It is necessary to attract external funding from charitable and environmental organizations, as well as local government bodies; to conduct systematic training courses, workshops, and seminars to enhance the digital competence of academic staff; to develop methodological guidelines for integrating AI technologies into the educational process; to establish unified standardized protocols for processing educational data; and to ensure a high level of cybersecurity within the virtual educational environment.

Future research should focus on scaling the IAEPMF experiment to the national level and conducting multi-institutional comparative analyses of AI model performance in diverse HEI contexts. Long-term impact assessment of AI-based environmental training on graduates' professional competencies, as well as in-depth examination of ethical, economic, and equity dimensions of AI integration in higher education, represent important directions for advancing this field.

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6. Declarations

6.1. Ethical considerations

The pedagogical experiment was conducted in accordance with the internal norms and regulations for educational research adopted by higher education institutions in Ukraine. Informed consent for participation and personal data processing was obtained from all participants or their legal representatives. As the study involves minimal risk and does not interfere with the educational process or include medical procedures, it did not require formal ethics committee (IRB) approval. This exemption

aligns with the current recommendations of the Ministry of Education and Science of Ukraine and the internal policies of the participating universities regarding pedagogical and socio-humanitarian research.

6.2. Use of artificial intelligence (AI)

The authors declare that no generative artificial intelligence (AI) tools were used in the preparation, analysis, or writing of this manuscript.

6.3. Conflict of interest

The authors declare no conflicts of interest that could influence the objectivity or integrity of this research.

6.4. Funding

This research received no external funding.

References

Abulibdeh, A., Zaidan, E., & Abulibdeh, R. (2024). Navigating the confluence of artificial intelligence and education for sustainable development in the era of Industry 4.0: Challenges, opportunities, and ethical dimensions. *Journal of Cleaner Production, 437*, 140527. https://doi.org/10.1016/j.jclepro.2023.140527

Adiguzel, T., Kaya, M. H., & Cansu, F. K. (2023). Revolutionizing education with AI: Exploring the transformative potential of ChatGPT. *Contemporary Educational Technology, 15*(3), e429. https://doi.org/10.30935/cedtech/13152

Akour, M., & Alenezi, M. (2022). Higher education future in the era of digital transformation. *Education Sciences, 12*(11), 784. https://doi.org/10.3390/educsci12110784

Alamri, H. A., Watson, S., & Watson, W. (2021). Learning technology models that support personalization within blended learning environments in higher education. *TechTrends, 65*(1), 62–78. https://doi.org/10.1007/s11528-020-00530-3

Alotaibi, E., & Nassif, N. (2024). Artificial intelligence in environmental monitoring: In-depth analysis. *Discover Artificial Intelligence, 4*(1), 84. https://doi.org/10.1007/s44163-024-00198-1

Bibri, S. E. (2020). Data-driven environmental solutions for smart sustainable cities: Strategies and pathways for energy efficiency and pollution reduction. *Euro-Mediterranean Journal for Environmental Integration, 5*, 66. https://doi.org/10.1007/s41207-020-00211-w

Cherep, A., Voronkova, V., Cherep, O., & Nikitenko, V. (2024). Artificial intelligence as a tool for ensuring environmental safety and sustainable development. In S. Tabachnikov (Ed.), *Ukrainian science and education in the conditions of European integration* (pp. 39–54). GS Publishing Services. https://doi.org/10.51587/9798-9895-14670-2024-020-39-54

Costanza, R., Cumberland, J. H., Daly, H., Goodland, R., Norgaard, R., Kubiszewski, I., & Franco, C. (2014). *An introduction to ecological economics* (2nd ed.). CRC Press.

Goralski, M. A., & Tan, T. K. (2020). Artificial intelligence and sustainable development. *The International Journal of Management Education, 18*(1), 100330. https://doi.org/10.1016/j.ijme.2019.100330

Guo, Z., Huang, L., Yan, J., Zhang, H., Jia, X., Li, M., & Li, H. (2025). Artificial intelligence technology in environmental research and health: Development and prospects. *Environment International*, 109788. https://doi.org/10.1016/j.envint.2025.109788

Hino, M., Benami, E., & Brooks, N. (2018). Machine learning for environmental monitoring. *Nature Sustainability, 1*(10), 583–588. https://doi.org/10.1038/s41893-018-0142-9

Jauhiainen, J. S., & Guerra, A. G. (2023). Generative AI and ChatGPT in school children's education: Evidence from a school lesson. *Sustainability, 15*(18), 14025. https://doi.org/10.3390/su151814025

Leal Filho, W., Kim, E., Borsatto, J. M. L. S., & Marcolin, C. B. (2025). Using artificial intelligence in sustainability teaching and learning. *Environmental Sciences Europe, 37*(1), 124. https://doi.org/10.1186/s12302-025-01159-w

Li, X., Xiang, H., Zhou, X., & Jing, H. (2023, June). An empirical study on designing STEM+ AI teaching to cultivate primary school students' computational thinking perspective. In *Proceedings of the 2023 8th International Conference on Distance Education and Learning* (pp. 1–7). https://doi.org/10.1145/3606094.3606130

Mehdipour, F., Pashna, M., & Mahanti, A. (2018, April). A 3-tier solution for facilitating STEM education in primary schools. In *2018 International Conference on Learning and Teaching in Computing and Engineering (LaTICE)* (pp. 1–5). IEEE. https://doi.org/10.1109/LaTICE.2018.00-15

Morze, N., Varchenko-Trotsenko, L., Terletska, T., & Smyrnova-Trybulska, E. (2023). Artificial intelligence as primary school teacher assistant. *Open Educational E-Environment of Modern University, 15*, 97–115. https://doi.org/10.28925/2414-0325.2023.158

Mullainathan, S., & Spiess, J. (2017). Machine learning: An applied econometric approach. *Journal of Economic Perspectives, 31*(2), 87–106. https://doi.org/10.1257/jep.31.2.87

Oleksenko, R., Nikitenko, V., Voronkova, V., Bakka, T., Popova, A., Bohomaz, O., Zahrebelna, N., Afonov, R., Plakhotniuk, H., Vdovenko, V., & Bilohur, V. (2025). *AI tools in building sustainable education*. International Journal of Basic and Applied Sciences, 14(4), 508–514. https://doi.org/10.14419/2v1gkx25

Santos, M. R., & Carvalho, L. C. (2025). AI-driven participatory environmental management: Innovations, applications, and future prospects. *Journal of Environmental Management, 373*, 123864. https://doi.org/10.1016/j.jenvman.2024.123864

Singh, N., Gunjan, V. K., Mishra, A. K., Mishra, R. K., & Nawaz, N. (2022). SeisTutor: A custom-tailored intelligent tutoring system and sustainable education. *Sustainability, 14*(7), 4167. https://doi.org/10.3390/su14074167

Toderas, M. (2025). Artificial intelligence for sustainability: A systematic review and critical analysis of AI applications, challenges, and future directions. *Sustainability, 17*(17), 8049. https://doi.org/10.3390/su17178049

Vinuesa, R., Azizpour, H., Leite, I., Balaam, M., Dignum, V., Domisch, S., Felländer, A., Langhans, S. D., Tegmark, M., & Fuso Nerini, F. (2020). The role of artificial intelligence in achieving the Sustainable Development Goals. *Nature Communications, 11*(1), 233. https://doi.org/10.1038/s41467-019-14108-y

Wu, Y., Xu, M., & Liu, S. (2024). Generative artificial intelligence: A new engine for advancing environmental science and engineering. *Environmental Science & Technology, 58*(40), 17524–17528. https://doi.org/10.1021/acs.est.4c07216

Zanevych, Y., Yovbak, V., Basystiuk, O., Shakhovska, N., Fedushko, S., & Argyroudis, S. (2024). Evaluation of pothole detection performance using deep learning models under low-light conditions. *Sustainability, 16*(24), 10964. https://doi.org/10.3390/su162410964

Zhu, J. J., Jiang, J., Yang, M., & Ren, Z. J. (2023). ChatGPT and environmental research. *Environmental Science & Technology, 57*(46), 17667–17670. https://doi.org/10.1021/acs.est.3c01818